

Reproduction Guide

Minimum package required for an independent rerun

Current status

The manuscript documents the model and its reported results. Independent reproducibility is not complete until executable code and exact inputs are deposited. Do not mark the repository “fully reproducible” until every item below is present and tested on a clean machine.

Required deposits

- Exact source code used for background integration, including both DOP853 and Radau configurations.
- Modified CAMB source files or a patch against a named CAMB release/commit.
- Locked parameter file containing every cosmological, scalar-potential, transfer, transition, and numerical parameter.
- Environment file: operating system, Python version, compiler, CAMB version, packages, and package versions.
- Input-data manifest identifying every public dataset, release, file name, checksum, license, and official source.
- Scripts that generate each table and figure from raw outputs.
- Expected machine-readable outputs and SHA-256 checksums.
- A one-command or numbered clean-room reproduction sequence.

Recommended run order

1. Create the documented environment.
2. Run the contracting and bouncing background with constraint monitoring.
3. Run the independent stiff solver and compare the trajectory.
4. Extract and freeze the expanding $H(z)$ history.
5. Build the named CAMB version with the supplied patch.
6. Run the matched control and locked model from frozen parameter files.
7. Generate primary CMB, acoustic, BAO, growth, and lensing outputs.
8. Calculate statistics from the supplied covariance/window files.
9. Compare output hashes and numerical tolerances with the expected-results manifest.

Acceptance checks

- No division by H is used at the bounce.
- Constraint residual remains within the stated tolerance.
- Total transferred energy balances between scalar and radiation sectors.
- DOP853 and Radau agree within stated tolerances.
- $E(0)=1$ after locked normalization.
- Reported headline values are reproduced without manual retuning.
- All plots and tables are generated from scripts, not hand-entered numbers.